Exercise - 2.10

Determine the region in the plane determined by the inequalities:

(1) $x \leq 3y, \quad x \geq y.$
(2) $y \geq 2x, \quad -2x + 3y \leq 6.$
(3) $3x + 5y \geq 45, x \geq 0, y \geq 0.$
(4) $2x + 3y \leq 35, y \geq 2, x \geq 5.$
(5) $2x + 3y \leq 6,\ x + 4y \leq 4,\ x \geq 0, y \geq 0.$
(6) $x - 2y \geq 0,\ 2x - y \leq -2,\ x \geq 0, y \geq 0.$
(7) $2x + y \geq 8, x + 2y \geq 8, x + y \leq 6.$

2.8 Exponents and Radicals

First we shall consider exponents.

2.8.1 Exponents

Let $n \in \mathbb{N}$, $a \in \mathbb{R}$. Then $a^n = a \cdot a \cdots a$ (n times). If m is a negative integer and the real number $a \neq 0$, then $a^m = \dfrac{1}{a^{-m}}$.

Note that for any $a \neq 0$, we have $\dfrac{a}{a} = a^{1-1} = a^0 = 1$. It is also easy to see the following properties.

Properties of Exponents

(i) For $m, n \in \mathbb{Z}$ and $a \neq 0$, we have $a^m a^n = a^{m+n}$.
(ii) For $m, n \in \mathbb{Z}$ and $a \neq 0$, we have $\dfrac{a^m}{a^n} = a^{m-n}$.

2.8.2 Radicals

Question:

For $a \neq 0$ and $r \in \mathbb{Q}$, is it possible to define a^r?

First let us consider the case when $r = \dfrac{1}{n}$, $n \in \mathbb{N}$. Suppose there is a real number $y \in \mathbb{R}$ such that $y = a^{\frac{1}{n}}$. Then we must have $y^n = a$.

This problem is basically to finding inverse function of $y = x^n$. In order to understand better let us consider the graphs of the following functions:

(i) $f(x) = x^{2n}$, $n \in \mathbb{N}$ (ii) $g(x) = x^{2n+1}$, $n \in \mathbb{N}$



